

Transfer Learning for Freshness Classification of Fruits and Vegetables: A Comparative Study of Backbone Architectures and Training Conditions

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Abstract—Automated freshness classification of fruits and vegetables is a critical task for food safety, supply chain management, and retail quality control. This paper presents a systematic comparative study of transfer learning applied to this problem, evaluating three convolutional neural network backbone architectures — ResNet-18, EfficientNet-B0, and EfficientNet-B2 — under four training conditions that vary the degree of backbone fine-tuning and the presence of a non-linear projection head. Experiments are conducted on six publicly available datasets comprising over 55,000 images across multiple fruit and vegetable categories, including binary (fresh/rotten) and three-class (fresh/formalin/rotten) annotation schemes. We find that EfficientNet-B0 with a frozen backbone and projection head achieves the best performance on datasets with more than 1,500 images, reaching 100% macro F1-score on one benchmark, while ResNet-18 remains superior for very small datasets (<500 images). A key finding is that the optimal training condition differs by backbone family: fine-tuning the last residual block is beneficial for ResNet-18 (C4), but detrimental for EfficientNet variants, which perform best with a completely frozen backbone (C3). We further evaluate domain shift by testing trained models on a small collection of 121 real-world photos taken under natural lighting conditions, revealing a significant performance gap of up to 34.3 percentage points between studio-quality training data and real-world evaluation images.

Index Terms—transfer learning, freshness classification, convolutional neural networks, EfficientNet, ResNet, domain shift, food quality inspection, fine-tuning

I. INTRODUCTION

Food freshness is a critical quality attribute that directly affects consumer health, retail profitability, and supply chain efficiency. Manual inspection of fruits and vegetables is time-consuming, subjective, and difficult to scale to the volumes required by modern distribution systems. Automated computer vision systems offer a scalable and objective alternative, capable of processing large volumes of produce at low cost.

Recent advances in deep learning, particularly convolutional neural networks (CNNs) pre-trained on large-scale image datasets, have demonstrated strong performance on a wide range of visual classification tasks. Transfer learning [1] leverages these pre-trained representations to build effective classifiers

from limited domain-specific data — an important property for food quality applications where labeled datasets are expensive to collect.

Despite the growing body of work on food classification and freshness detection, a systematic comparison of backbone architectures and fine-tuning strategies under controlled conditions remains lacking. Furthermore, the domain shift between publicly available dataset images (typically acquired in controlled photographic conditions) and real-world images captured by mobile devices has received limited attention.

This paper addresses both gaps through the following contributions:

- A controlled comparison of three CNN backbone architectures (ResNet-18, EfficientNet-B0, EfficientNet-B2) and four training conditions (C1–C4) across six benchmark datasets totalling 55,040 images.
- The identification of a consistent interaction between backbone family and optimal training condition: C4 (partial fine-tuning + projection head) for ResNet-18 vs. C3 (frozen backbone + projection head) for EfficientNet variants.
- A quantitative domain shift analysis showing that models trained on internet-sourced studio photos can lose up to 34 percentage points of F1-score when evaluated on real-world mobile camera images.
- An open-source pipeline implementing the full experimental workflow from data preparation to evaluation reporting.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the methodology. Section IV presents experimental results. Section V discusses key findings. Section VI concludes the paper.

II. RELATED WORK

A. Food Quality Classification with Deep Learning

Deep learning has been widely applied to food recognition and quality assessment [?]. Several works specifically target freshness detection using CNNs trained from scratch or with

transfer learning [?]. Bangladeshi et al. evaluated multiple deep learning architectures for fresh/rotten detection of fruits and vegetables, finding that pre-trained models significantly outperform models trained from scratch [?].

B. Backbone Architectures

ResNet [2] introduced skip connections to address the vanishing gradient problem, enabling deeper networks that generalize well across domains. EfficientNet [3] proposed compound scaling — simultaneously adjusting network depth, width, and input resolution — achieving superior ImageNet accuracy with fewer parameters than comparably-performing ResNet variants. EfficientNet-B0 and B2 achieve 77.1% and 80.1% top-1 accuracy on ImageNet respectively, compared to 69.8% for ResNet-18, with approximately half the parameters.

C. Transfer Learning Strategies

The most common transfer learning strategies for image classification range from linear probing (training only the final classifier) to full fine-tuning (updating all weights) [1]. Kornblith et al. [4] showed that better ImageNet features transfer better across tasks. The intermediate strategy of fine-tuning only the later layers — which capture more task-specific features — is widely adopted as a balance between adaptation and catastrophic forgetting [5].

III. METHODOLOGY

A. Datasets

Six publicly available datasets are used in this study (Table I). Together they cover 55,040 images across fruits, vegetables, and citrus varieties, with binary freshness labels (fresh/rotten) in five datasets and a three-class annotation (fresh/formalin/rotten) in one. The FruitVision dataset is particularly relevant as the formalin class presents a significant challenge: chemically-treated samples are visually similar to fresh specimens, analogous to the difficulty of detecting subtle spoilage in fermented foods.

TABLE I: Datasets used in this study. [†]Three-class annotation.

Dataset	ID	Images	Classes	Source
FruitVision [†]	MFV	10,154	3	Mendeley
Fruits Classification	MFR	1,655	2	Mendeley
Lemon Varieties	MLM	1,956	2	Mendeley
Fruits Quality	KFQ	359	2	Kaggle
Fruits Fresh/Rotten	KFR	13,599	2	Kaggle
Fresh & Stale	KFS	27,317	2	Kaggle
Total		55,040		

B. Data Preprocessing

All images are resized so the shortest side is 256 pixels and then center-cropped to 224×224 for validation. Training images undergo the following augmentation pipeline: random resized crop to 224×224 , random horizontal flip, $\pm 30^\circ$ rotation, color jitter (brightness, contrast, saturation, hue), and Gaussian blur (kernel size 3). All images are normalized with ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). A stratified 80/20 train/validation split with fixed random seed 42 is applied to all datasets.

C. Backbone Architectures

Three pre-trained CNN backbones are evaluated, all initialized with ImageNet weights from the `torchvision` library:

- **ResNet-18** [2]: 11.2M parameters, 512-dimensional feature output from global average pooling.
- **EfficientNet-B0** [3]: 5.3M parameters, 1280-dimensional feature output.
- **EfficientNet-B2** [3]: 9.1M parameters, 1408-dimensional feature output.

D. Training Conditions

Four training conditions form an ablation study over backbone fine-tuning and classification head architecture (Table II).

TABLE II: Training conditions.

ID	Backbone	Head	Description
C1	Frozen	Linear only	Linear probe
C2	Layer4 free	Linear only	Partial fine-tuning
C3	Frozen	Proj. + Linear	Full head, frozen backbone
C4	Layer4 free	Proj. + Linear	Full head + fine-tuning

In conditions C3 and C4, a two-layer projection head maps the backbone output to a 128-dimensional embedding before classification: $d_{\text{out}} \xrightarrow{\text{FC}} 256 \xrightarrow{\text{ReLU}} \xrightarrow{\text{FC}} 128 \xrightarrow{\text{ReLU}} C$, where C is the number of classes. In condition C4, `layer4` of the backbone is unfrozen and trained with a reduced learning rate $\text{lr}_{\text{backbone}} = 10^{-5}$, compared to $\text{lr} = 10^{-3}$ for the head, to mitigate catastrophic forgetting [5].

All models are trained for 60 epochs using the Adam optimizer with cosine annealing learning rate schedule and weight decay 10^{-4} . Batch size is 32.

E. Evaluation Metrics

Given the potential class imbalance in food datasets, accuracy alone is insufficient. Let TP_c , FP_c , FN_c denote true positives, false positives, and false negatives for class c , and C the number of classes. We report:

$$\overline{F1} = \frac{1}{C} \sum_{c=1}^C \frac{2TP_c}{2TP_c + FP_c + FN_c} \quad (1)$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (2)$$

Additionally, we report Area Under the ROC Curve (AUC-ROC) and per-class F1, precision, and recall. For food safety applications, recall on the *rotten* class is particularly critical: a false negative (rotten food classified as fresh) poses a direct health risk.

IV. EXPERIMENTS

A. Phase 1: ResNet-18 Baseline

Table III reports the best macro F1-score achieved by ResNet-18 for each dataset and condition (state mode, 60 epochs).

Condition C4 achieves the best result in four of six datasets; C2 wins on the two datasets where the number of training samples is smallest (KFQ: 287 train samples; MFV: 8,123 train samples but three-class complexity).

TABLE III: Best macro F1-score (%) — ResNet-18, state mode. Bold: best condition per dataset.

Dataset	C1	C2	C3	C4	Best
KFR	97.6	99.8	99.4	99.8	C2/C4
KFS	95.0	98.9	98.0	99.0	C4
MLM	96.2	98.0	96.2	98.5	C4
MFR	94.3	95.5	95.2	96.4	C4
MFV	78.3	91.0	86.7	90.6	C2
KFQ	86.1	91.6	86.0	88.9	C2

B. Phase 2: Backbone Comparison

Table IV compares the best macro F1-score per dataset across the three backbone architectures (state mode, best condition). EfficientNet-B0 with condition C3 achieves the best result in four of six datasets. ResNet-18 wins only on KFQ (359 images), the smallest dataset in the benchmark.

TABLE IV: Best macro F1-score (%) by backbone architecture, state mode. Bold: overall best per dataset. *Experiment in progress.

Dataset	ResNet-18		EffNet-B0		EffNet-B2	
	F1	Cond	F1	Cond	F1	Cond
KFR	99.8	C4	100.0	C3	99.9	C3
KFS	99.0	C4	99.4	C3	*	—
MLM	98.5	C4	98.7	C3	—	—
MFR	96.4	C4	97.0	C3	—	—
MFV	91.0	C2	91.6	C3	92.3	C3
KFQ	91.6	C2	86.1	C4	—	—

C. Domain Shift Analysis

To measure generalization to real-world conditions, trained models are evaluated on *own_dataset*: 121 strawberry photos taken with a mobile phone camera under natural indoor lighting. None of the training datasets contain photos of strawberries taken in these conditions. Table V reports the top results sorted by evaluation F1-score, with the domain shift (Drop = $F1_{\text{train}} - F1_{\text{eval}}$) for each model.

TABLE V: Cross-dataset evaluation on own photos (state mode).

Model ID	$F1_{\text{train}}$	$F1_{\text{eval}}$	Drop
MLM-R18-C4	98.5	91.7	6.8
MLM-EB0-C4	95.4	91.7	3.7
MLM-R18-C2	98.0	87.5	10.5
MFR-EB0-C3	97.0	87.3	9.7
KFS-R18-C4	99.0	87.3	11.7
KFR-R18-C4	99.8	70.8	29.0
KFR-EB0-C3	100.0	65.7	34.3

V. RESULTS AND DISCUSSION

Finding 1: C3 is optimal for EfficientNet; C4 for ResNet-18. In four of five completed datasets, EfficientNet-B0 achieves its best performance under C3 (frozen backbone + projection head), while ResNet-18 consistently benefits from partial fine-tuning (C4). We attribute this to the richer pre-trained features of EfficientNet: its 1280-dimensional output already captures discriminative cues for freshness without backbone adaptation, whereas ResNet-18’s 512-dimensional output benefits from specialized high-level feature adaptation. Fine-tuning EfficientNet’s backbone with the available data sizes appears to introduce overfitting rather than beneficial specialization.

Finding 2: Dataset size determines the winning backbone. Below approximately 500 training samples, ResNet-18 outperforms EfficientNet-B0 by up to 5.5 percentage points (KFQ dataset: 91.6% vs. 86.1%). Above 1,500 training samples, EfficientNet-B0 consistently matches or surpasses ResNet-18. This is consistent with the larger head parameters of EfficientNet requiring more data to generalize: a $1280 \rightarrow 256 \rightarrow 128 \rightarrow C$ projection head has approximately $3\times$ more parameters than the equivalent ResNet-18 head.

Finding 3: EfficientNet-B0 achieves perfect classification on KFR. On the Fruits Fresh/Rotten dataset (apple, banana, orange), EfficientNet-B0 under C3 achieves 100% macro F1-score, MCC = 1.0, and AUC-ROC = 1.0. This represents a perfect separation between fresh and rotten classes, suggesting that the visual differences in this dataset are highly discriminative and well-captured by EfficientNet’s pre-trained features without backbone adaptation.

Finding 4: Formalin represents a fundamental classification challenge. FruitVision is the only dataset where no backbone exceeds 92.3% F1-score, despite having 10,154 images. The formalin class is chemically treated to appear visually similar to fresh samples, creating a subtle spectral shift that is difficult to detect from RGB images alone. EfficientNet-B2 achieves the best result (92.3%), suggesting that larger, more expressive backbones provide marginal improvements on hard discrimination tasks. This finding is directly relevant to the detection of subtle spoilage in fermented foods.

Finding 5: High training F1 does not imply good generalization. The Fruits Fresh/Rotten model that achieves 100% F1 on its own training domain (KFR-EB0-C3) performs worst in cross-dataset evaluation, dropping to 65.7% on real photos (drop of 34.3 points). Conversely, the Lemon Varieties model (MLM-R18-C4, 98.5% F1) achieves the highest evaluation performance (91.7%, drop of only 6.8 points). We attribute this inversion to the diverse photographic conditions in the Lemon Varieties dataset, which expose the model to a wider range of lighting and background variation during training.

Finding 6: Domain shift is quantifiable and large. Across all evaluated models, the mean domain shift is 16.0 percentage points, with a range from 3.7 to 34.3 points. Models trained on datasets with more uniform photographic conditions (KFR: studio photos with white backgrounds) exhibit the largest

shift. This strongly motivates the collection of domain-specific datasets that match real deployment conditions when high-accuracy production systems are required.

VI. CONCLUSION

This paper presented a systematic comparison of transfer learning strategies for fruit and vegetable freshness classification. Three backbone architectures and four training conditions were evaluated across six benchmark datasets totalling 55,040 images.

The key findings are: (1) EfficientNet-B0 with a frozen backbone and projection head (C3) achieves the best performance on medium-to-large datasets, including a perfect 100% F1-score on one benchmark; (2) ResNet-18 with layer4 fine-tuning (C4) remains the best choice for datasets with fewer than 500 training samples; (3) domain shift between studio-quality training data and real-world mobile photos ranges from 3.7 to 34.3 percentage points, motivating the collection of domain-specific labeled data.

Future work will extend this pipeline to Korean food categories (kimchi, rice cake, fermented side dishes), where the visual similarity between fresh and fermented states is analogous to the formalin challenge identified in this study.

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